

Java Collections Source Files

1. LinkedHashSetDemo.java

```
import java.util.LinkedHashSet;

public class LinkedHashSetDemo {

    public static void main(String[] args) {

        LinkedHashSet set = new LinkedHashSet<>();

        set.add("Ali");
        set.add("Sara");
        set.add("Aya");
        set.add("Ali");

        System.out.println(set);

        set.remove("Sara");

        System.out.println(set);

        System.out.println(set.contains("Aya"));
    }
}
```

2. LinkedListDemo.java

```
import java.util.LinkedList;

public class LinkedListDemo {
    public static void main(String[] args){
        LinkedList list = new LinkedList<>();

        // add last
        list.add("Aya");
        list.add("Islam");
        list.add("Omar");

        System.out.println(list);

        // add first
        list.addFirst("Ali");
        System.out.println(list);

        // add middle
        list.add(1, "Ibrahim");
        System.out.println(list);

        // get
        System.out.println(list.get(2));
        System.out.println(list.getFirst());
        System.out.println(list.getLast());

        // remove
        list.removeFirst();
        System.out.println(list);

        list.removeLast();
        System.out.println(list);

        list.remove(1);
        System.out.println(list);

        // update
        list.set(1, "Malek");
        System.out.println(list);

        // search
        System.out.println(list.contains("aya"));
        System.out.println(list.contains("Malek"));

    }
}
```

3. PriorityQueueDemo.java

```
import java.util.PriorityQueue;

public class PriorityQueueDemo {
    public static void main (String[] args){
        PriorityQueue pq = new PriorityQueue<>();

        pq.offer(40);
        pq.offer(10);
        pq.offer(30);
        pq.offer(20);

        System.out.println(pq);
        System.out.println(pq.peek());

        pq.poll();
        System.out.println(pq);
    }
}
```

4. StackDemo.java

```
import java.util.Stack;

public class StackDemo {
    public static void main (String [] args){

        Stack stack = new Stack<>();

        // add
        stack.push(1);
        stack.push(2);
        stack.push(3);
        stack.push(4);
        stack.push(5);
        stack.push(100);

        System.out.println(stack);

        // get top element
        System.out.println(stack.peek());

        // update
        stack.set(2,5555);
        System.out.println(stack);

        // search
        System.out.println(stack.contains(100));
        System.out.println(stack.contains(1513));

        // remove
        stack.pop();
        System.out.println(stack);

        stack.pop();
        System.out.println(stack);

    }
}
```

5. VectorDemo.java

```
import java.util.Vector;

public class VectorDemo {
    public static void main(String[] args){
        Vector v = new Vector<>();

        // add last
        v.add(2);
        v.add(10);
        v.add(18);
        System.out.println(v);

        // add first
        v.addFirst(100);
        System.out.println(v);

        // add middle
        v.add(1,159);
        System.out.println(v);

        // get
        System.out.println(v.get(2));
        System.out.println(v.getFirst());
        System.out.println(v.getLast());

        // remove
        v.removeFirst();
        System.out.println(v);

        v.removeLast();
        System.out.println(v);

        v.remove(1);
        System.out.println(v);

        v.removeElement(10);
        System.out.println(v);

        // update
        v.set(0,888);
        System.out.println(v);

        // search
        System.out.println(v.contains(100));
        System.out.println(v.contains(888));

    }
}
```

6. ArrayDequeDemo.java

```
import java.util.ArrayDeque;

public class ArrayDequeDemo {

    public static void main(String[] args) {

        ArrayDeque deque = new ArrayDeque<>();

        // Add
        deque.addFirst("B");
        deque.addLast("C");
        deque.addFirst("A");

        System.out.println(deque);

        // Peek
        System.out.println(deque.peekFirst());
        System.out.println(deque.peekLast());

        // Remove
        deque.removeFirst();
        deque.removeLast();

        System.out.println(deque);

        // Stack
        deque.push("X");
        deque.push("Y");

        System.out.println(deque);

        System.out.println(deque.pop());
        System.out.println(deque);

        // queue
        deque.offer("Z");
        System.out.println(deque);

        deque.offerFirst("A");
        deque.offerLast("B");
        System.out.println(deque);

        deque.poll();
        System.out.println(deque);

        deque.pollLast();
        System.out.println(deque);

    }
}
```

7. ArrayListDemo.java

```
import java.util.ArrayList;

public class ArrayListDemo {

    public static void main(String[] args){
        ArrayList list = new ArrayList<>();

        // add last
        list.add("Aya");
        list.add("Ali");
        list.add("Omar");

        System.out.println(list);

        // add first
        list.addFirst("Muhammad");
        System.out.println(list);

        // add middle
        list.add(1, "Ahmad");
        list.add(3, "Ali");
        System.out.println(list);

        // get
        System.out.println(list.get(3));

        // update
        list.set(3, "Islam");
        System.out.println(list);

        // search
        System.out.println(list.contains("Aya"));
        System.out.println(list.contains("Duaa"));

        // remove by index
        list.remove(1);
        System.out.println(list);

        // remove by element
        list.remove("Islam");
        System.out.println(list);
    }
}
```

8. HashSetDemo.java

```
import java.util.HashSet;

public class HashSetDemo {

    public static void main(String[] args) {

        HashSet set = new HashSet<>();

        set.add("Ali");
        set.add("Aya");
        set.add("Ali");

        System.out.println(set);

        System.out.println(set.contains("Aya"));

        set.remove("Aya");

        System.out.println(set);

    }
}
```